Sat-Link: An Open-Source Simulator for ITU-R-Compliant Satellite Link-Budget Analysis

Nikolaos Kalaitzakis
Department of Electrical & Computer Engineering (ECE)
Hellenic Mediterranean University
Heraklion, Crete, Greece
th20458@edu.hmu.gr

Ioannis Papoudakis
Department of Electrical & Computer Engineering (ECE)
Hellenic Mediterranean University
Heraklion, Crete, Greece
th20642@edu.hmu.gr

Alexandros Kontos
Department of Electrical & Computer Engineering (ECE)
Hellenic Mediterranean University
Heraklion, Crete, Greece
th20308@edu.hmu.gr

Dimitrios I. Stratakis
Department of Electrical & Computer Engineering (ECE)
Hellenic Mediterranean University
Heraklion, Crete, Greece
dstrat@hmu.gr

Spyridon Panagiotakis
Department of Electrical & Computer Engineering (ECE)
Hellenic Mediterranean University
Heraklion, Crete, Greece
spanag@hmu.gr

Manolis Tampouratzis
Department of Electrical & Computer Engineering (ECE)
Hellenic Mediterranean University
Heraklion, Crete, Greece
tampouratzis@hmu.gr

***Abstract*—Sat-Link is a desktop-based, open-source simulator which models a complete end-to-end satellite link budget using ITU-R Recommendations. The model uses ITU-R methodology to model free-space path loss (Friis), atmospheric gaseous attenuation, rain attenuation, scintillation, cloud attenuation, and rain height. Sat-Link also supports integrated orbit propagation using VLA with Keplerian Kinematics, J2 secular perturbation with optional drag parameterization, for initial assessments of low-altitude "what-if" scenarios. The modular structure of Sat-Link is built using Python/PyQt5, providing seven independent and interactive tabs that include visualization tools, as well as a full analysis dialog for performing statistical, spectral, and regulatory analysis. The behavior of high-power amplifiers (HPAs) is determined using cascaded noise analyses utilising Friis' formula. In addition, two non-linear models (Saleh and Rapp) are used to calculate output back-off and intermodulation for composite transmitted signals compared to the total carrier-to-noise density ratio (C/N0). Three independent scenarios were validated against appropriate benchmarks yielding a mean absolute error of C/N0 of 0.23 dB, with a maximum of 0.30 dB. Therefore, Sat-Link is an ideal tool for teaching in the classroom, and for early phases of system trade studies, including parametric sweeps and other sensitivity analyses for design space exploration. A full mission qualification will be outside the scope of the simulator.**

***Index Terms*—Satellite communication systems, link budget, ITU-R, propagation modeling, transponder nonlinearity, educational software.**

I. Introduction

Satellite link budget analysis is one of the most important topics in the satellite communications curriculum and at the beginning of the preliminary mission design process [1], [2]. However, in many educational pathways, the students rely upon static spreadsheets or commercial software that have limited algorithmic transparency to support their designs based on decisions made during the design phase (orbit, frequency, coding, etc.) and weather, to name but a few [3]–[5].Sat-Link provides a solution to this problem with an open source desktop application which has an interactive interface with all of the necessary tools to provide a complete ITU-R aligned uplink and downlink calculation chain, along with transponder nonlinearities and orbit dependent geometric effects [6]–[14]. The Sat-Link simulator can be used to model LEO, MEO, GEO, and HEO scenarios and allows users to run both real time visual diagnostic and offline statistical studies [15].These are the primary contributions provided by the Sat-Link project.First, the Sat-Link project developed a modular Qt5 graphical user interface with seven different visualization windows that allow users to inspect their link budgets, view the behavior of their bit error rates (BER) due to signal degradation [3], [4], observe the quality of their constellations, observe the ground tracks of the satellites, visualize the orbits of the satellites in three dimensions, create animations of the links between the satellites and Earth stations and the behavior of their transponders.Second, the Sat-Link project also developed an ITU-R aligned link budget and transponder chain that includes a Friis noise cascade with Saleh and Rapp HPA models and intermodulation effects.Third, the Sat-Link project has developed a reproducible benchmark test set and has established clear scope limits for educational and early stage use. Overall, Sat-Link bridges the gap between theoretical link-budget analysis and practical system design by providing a transparent, reproducible, and ITU-R-aligned environment for education and early-stage satellite mission studies,and lastly for ham radio operators (i.e. licensed-users).

II. Related Work

Existing satellite-link tools span commercial platforms, numerical environments, and astrodynamics libraries. Furthermore, measured studies show strong regional sensitivity

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that makes transparent propagation modeling essential: for instance, Usman *et al.* reported 0.01% rain attenuation ranging from 14–22 dB (Ku) to 33–48 dB (Ka) depending on the elevation angle [5]. Four representative tools highlight current gaps:

- **STK [16]:** strong mission geometry, but communication-chain internals lack transparency for algorithm teaching.
- **MATLAB:** strong PHY experimentation, but complete ITU-R workflows require custom scripts in a commercial stack.
- **GMAT:** open and reliable orbit design, but lacks integrated atmospheric and transponder link-budget workflows.
- **Orekit:** excellent astrodynamics library, but leaves communication-specific GUIs and link-budget composition to the user.

This comparison clarifies Sat-Link's design goal: not to replace high-end mission suites, but to provide an auditable, standards-aligned, and classroom-ready environment connecting orbit, propagation, and transponder effects in one workflow.

III. Software Architecture

The simulator follows a model–view–controller (MVC) pattern implemented in Python 3.9 with PyQt5 for the graphical front-end. Fig. 1 illustrates the layered design.

A. Module Organisation

The implementation now follows the current repository layout:

- `main_advanced.py`: main desktop entry point.
- `models/`: link budget, orbit, modulation, beam pattern, satellite, transponder, and constants modules.
- `gui/visualizations/`: seven primary tab views (link budget, BER, constellation, ground track, 3D view, link diagram, transponder).
- `gui/advanced_analysis/`: advanced analysis dialog.
- `gui/hub/`, `gui/panels/`, `gui/dialogs/`, `gui/core/`: hub management, header/dashboard panels, dialogs, and shared UI/splash/canvas components.
- `gui/utils/`: export, preset, and save/load utilities.
- `assets/` and `txt_info/`: static resources and bundled reference text files.
- `requirements.txt`, `install.ps1`, `run_advanced.ps1`: dependency and Windows setup/run scripts.

Dependencies include PyQt5/PyQtWebEngine, NumPy, SciPy, Matplotlib, pyqtgraph, Plotly, Pandas, Seaborn, and optional PyVista/PyVistaQt for 3D rendering, with Skyfield/SGP4/pyorbital for orbital workflows. The ITU-R propagation models used in Sat-Link are implemented directly in the project's own Python modules, with local equations and embedded coefficient tables derived from the Recommendations; NumPy/SciPy are used for numerical operations and interpolation, rather than an external ITU-R Python package.

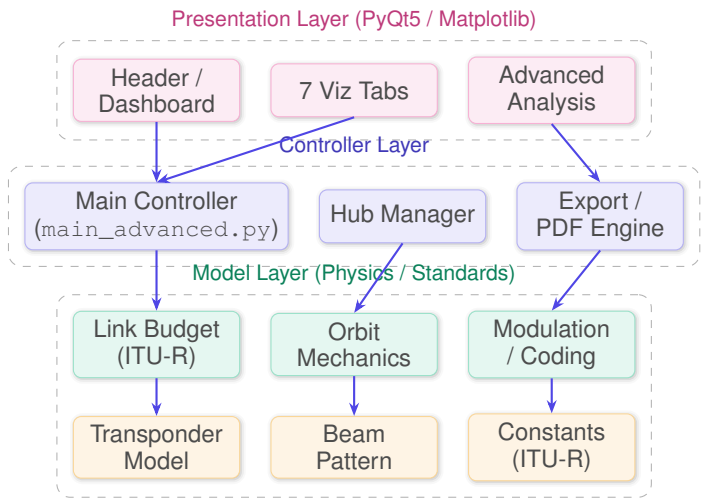

Fig. 1. Three-layer simulator architecture: presentation, controller, and model layers.

IV. Mathematical Models

A. Link-Budget Chain

The complete link budget follows the cascaded carrier-to-noise methodology of ITU-R S.1328 [2]. Each link direction (uplink and downlink) is computed independently, then combined at the transponder output. The key expressions are summarised below.

1) Free-Space Path Loss (ITU-R P.525):

$$\text{FSPL [dB]} = 20\log_{10}(d) + 20\log_{10}(f) + 92.45 \tag{1}$$

where d is the slant range in km and f the carrier frequency in GHz.

2) Rain Attenuation (ITU-R P.838-3 / P.618-14): The specific attenuation γ_R is:

$$\gamma_R = k\,R^{\alpha} \quad \text{[dB/km]} \tag{2}$$

where R is the rain rate in mm/h and the coefficients (k, α) are frequency- and polarisation-dependent as tabulated in ITU-R P.838-3 [7]. The total rain attenuation is:

$$A_{\text{rain}} = \gamma_R \cdot L_s \cdot r \tag{3}$$

with L_s the effective slant path length and r a reduction factor per [6].

3) Gaseous Attenuation (ITU-R P.676-12): Oxygen and water-vapour absorption are modelled as:

$$A_{\text{atm}} = \frac{\gamma_o\,h_o + \gamma_w\,h_w}{\sin E} \tag{4}$$

where γ_o, γ_w are specific attenuations, $h_o \approx 6\,\text{km}$, $h_w \approx 2\,\text{km}$ the equivalent heights, and E the elevation angle [8].

4) Cloud Attenuation (ITU-R P.840-8): Cloud and fog attenuation is computed from the integrated liquid water content L (kg/m^2) and a frequency-dependent specific attenuation coefficient K_l:

$$A_{\text{cloud}} = \frac{K_l \cdot L}{\sin E} \quad \text{[dB]} \tag{5}$$

where K_l depends on frequency and temperature per the Rayleigh scattering model [9].

5) Tropospheric Scintillation (ITU-R P.618-14): Amplitude scintillation is characterised by its standard deviation:

$$\sigma_{\text{scint}} = \sigma_{\text{ref}} \cdot f^{0.45} \cdot \frac{g(x)}{(\sin E)^{1.3}} \tag{6}$$

where σ_{ref} is a reference deviation dependent on N_{wet} (wet-term refractivity), f the frequency in GHz, and $g(x)$ an antenna-averaging function. The fade depth exceeded for a time percentage p is then $A_s(p) = a(p)\,\sigma_{\text{scint}}$. In the current implementation, operational plots use the common $p = 0.01\%$ approximation $A_s \approx 2.6\,\sigma_{\text{scint}}$.

6) Slant Range Geometry: The Earth–satellite slant range d relates to the elevation angle E and satellite altitude h above the Earth's surface:

$$d = R_E \left[\sqrt{\left(\frac{R_E + h}{R_E}\right)^2 - \cos^2 E} - \sin E \right] \tag{7}$$

where $R_E = 6371\,\text{km}$ is the mean Earth radius. This expression is used in both the FSPL and the atmospheric path-length calculations.

7) Carrier-to-Noise Density Ratio:

$$\frac{C}{N_0} = \text{EIRP} - \text{FSPL} - A_{\text{total}} + \frac{G}{T} - k_B \tag{8}$$

with $k_B = -228.6\,\text{dBW/K/Hz}$ the Boltzmann constant.

8) Total System C/N_0 (ITU-R S.1328): The total system performance combines uplink, downlink, and intermodulation:

$$\left(\frac{C}{N_0}\right)_{\text{total}}^{-1} = \left(\frac{C}{N_0}\right)_{\text{up}}^{-1} + \left(\frac{C}{N_0}\right)_{\text{dn}}^{-1} + \left(\frac{C}{IM_0}\right)^{-1} \tag{9}$$

B. Transponder Model

The satellite transponder is modelled as a four-stage cascaded chain (Fig. 3): low-noise amplifier (LNA), mixer, intermediate-frequency amplifier (IF Amp), and power amplifier (PA/HPA).

The system noise temperature follows Friis' formula:

$$T_{\text{sys}} = T_1 + \frac{T_2}{G_1} + \frac{T_3}{G_1 G_2} + \frac{T_4}{G_1 G_2 G_3} \tag{10}$$

where T_i and G_i denote the noise temperature and power gain of the i-th stage. Noise figure F and noise temperature are related by $T = T_0(F-1)$ with $T_0 = 290\,\text{K}$. The overall noise figure of the cascade is dominated by the first stage (LNA), motivating the use of cooled or ultra-low-noise front-ends in sensitive receivers.

The HPA nonlinearity is characterised by two models. The **Saleh model** [12] (TWTA) defines the AM/AM conversion as:

$$g(r) = \frac{\alpha_a\, r}{1 + \beta_a\, r^2} \tag{11}$$

where r is the normalised input amplitude and (α_a, β_a) are device-specific parameters. The **Rapp model** [13] (SSPA) uses:

$$g(r) = \frac{r}{\left[1 + (r/A_{\text{sat}})^{2p}\right]^{1/2p}} \tag{12}$$

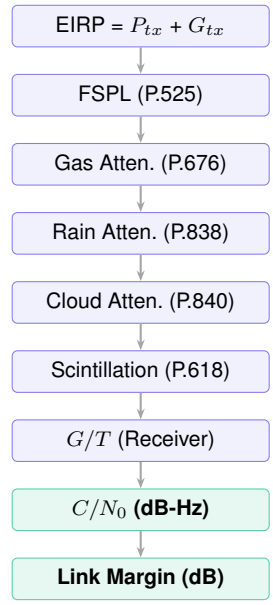

Fig. 2. Sequential computation flow of a single-link budget. Each step implements the referenced ITU-R Recommendation.

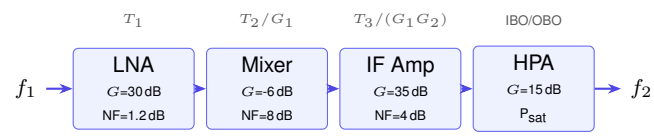

Fig. 3. Four-stage transponder cascade model with Friis' noise formula annotation. The HPA stage supports TWTA (Saleh), SSPA (Rapp), and linearised models.

with smoothness factor p ($p = 1$ yields soft clipping; $p \to \infty$ approaches ideal limiting). Input back-off (IBO) and output back-off (OBO) are derived from these models. The software then maps IBO to carrier-to-intermodulation ratio C/IM through an empirical model-family-dependent relation (TWTA/SSPA), and the resulting carrier-to-intermodulation ratio C/IM is fed into (9).

C. Orbital Mechanics

Satellite positions are propagated using Keplerian two-body mechanics augmented with J2 secular perturbations on the right ascension of the ascending node ($\dot{\Omega}$), argument of perigee ($\dot{\omega}$), and mean motion [14]:

$$\dot{\Omega} = -\frac{3}{2}\, n\, J_2 \left(\frac{R_E}{p}\right)^2 \cos i \tag{13}$$

where $p = a(1 - e^2)$ is the semi-latus rectum, n the mean motion, and $J_2 = 1.08263 \times 10^{-3}$. A first-order drag parameterisation using the US Standard Atmosphere 1976 density profile and ballistic coefficient $B = C_D A/m$ is included for preliminary LEO what-if analysis.

Ground-track computation employs ECI-to-ECEF rotation via the Greenwich Mean Sidereal Time (GMST), following the IAU-76/FK5 convention.

D. Modulation and BER Performance

Theoretical BER curves are implemented for seven modulation schemes: BPSK, QPSK, 8PSK, 16-QAM, 64-QAM, 16-APSK, and 32-APSK. For BPSK and QPSK (which share the same per-bit performance):

$$P_b = \frac{1}{2}\,\mathrm{erfc}\Big(\sqrt{E_b/N_0}\Big) \tag{14}$$

For rectangular M-QAM constellations the symbol error probability is approximated by:

$$P_s \approx 1 - \left[1 - \frac{2(\sqrt{M}-1)}{\sqrt{M}}\,\mathrm{erfc}\left(\sqrt{\frac{3\,E_s}{2(M-1)\,N_0}}\right)\right]^2 \tag{15}$$

with $E_s = E_b \log_2 M$ and Gray-coded bit error rate $P_b \approx P_s/\log_2 M$. Closed-form exact BER is used for BPSK/QPSK, while higher-order schemes are evaluated with standard analytical approximations [4].

Channel coding gains for LDPC, Turbo, and convolutional codes are applied per DVB-S2/S2X specifications [3]. The effective coding gain G_c (dB) shifts the BER curve horizontally, so the coded performance at rate r_c requires $E_b/N_0|_{\text{coded}} = E_b/N_0 - G_c$.

V. Graphical User Interface

The GUI is organised around a left-side control panel and a right-side tabbed visualisation area (Fig. 4). Its role is to expose model parameters and diagnostics, while all computation remains in the model layer described in Sections III–IV.

A. Operational Summary

The seven tabs are grouped into three functional blocks:

- **Link diagnostics:** link-budget terms, BER curves, and IQ constellation views.
- **Geometry views:** 2D ground track and interactive 3D orbit visualization.
- **Channel/transponder views:** animated link state and cascaded transponder behavior (including Saleh/Rapp HPA models).

Scenario management is handled by a hub workflow for defining satellites, ground stations, and presets, allowing rapid switching between candidate architectures during trade studies.

VI. Advanced Analysis Suite

The Advanced Analysis dialog extends the baseline tabs with grouped studies for: (i) deterministic parameter sweeps (elevation, frequency, rain rate), (ii) stochastic performance estimation (Monte Carlo and fade dynamics per ITU-R P.1623 [15]), (iii) spectral and interference checks, and (iv) regulatory sanity checks against PFD/EIRP masks and off-axis antenna references [10], [11]. This layer is intended for rapid trade-space exploration; where closed-form standards models are unavailable, reduced-order approximations are explicitly flagged in the interface.

VII. Validation and Results

The simulator was validated against three reference scenarios (Table I). For each case, the reference value was computed independently from textbook/standards equations (outside the GUI) using the same parameter set.

All instances yielded results within +/-0.3 dB agreement. The mean absolute error (MAE) was calculated at 0.23 dB and the root mean squared error (RMSE) was calculated at 0.24 dB, which is well within the typical +/-1 dB ITU-R modeling uncertainty band used for engineering link studies. Independent reference values were also generated using the corresponding textbook and standards-based equations. Finally, the rain attenuation calculation was cross-validated with the ITU-R P.838-3 coefficient tables, and there was no numerical disagreement between the rain attenuation calculation and the ITU-R P.838-3 coefficient tables for all frequency/polarization combinations from 1 to 100 GHz.

A. Error Characterisation

To make the validation outcome more interpretable for design decisions, Table II reports derived per-scenario error indicators. The worst-case absolute deviation is 0.3 dB (GEO/Ka rain case), while clear-sky GEO/Ku and LEO/S both remain at 0.2 dB.

Two observations follow. First, the signed residuals (-0.2, $+0.2$, -0.3 dB) indicate a small net negative bias (-0.1 dB mean signed error), i.e., slight conservative prediction of C/N_0. Second, error magnitude increases in the rain-loaded Ka-band case, which is consistent with the higher sensitivity of high-frequency links to compound atmospheric and nonlinear-transponder effects. Even in that case, the deviation remains significantly below the ± 1 dB uncertainty band commonly used in preliminary link design.

Tab 3 — IQ Constellation: Renders symbol constellations with configurable noise overlay and ideal reference symbol locations (Fig. 6).

Tab 6 — Link Diagram: A schematic diagram showing the dynamic path of the satellite to ground link, including dynamic weather effects such as clouds and rain drops, can be toggled by a clear sky / rainy condition toggle switch that will also update the budget calculations in real-time (Fig. 8).

B. Implications for Trade Studies

For early design exploration and education purposes, less than 0.3 dB agreement signifies that the ranking of possible configurations (i.e. aperture growth versus power growth) is unaffected by at least one-third of the tool's random numerical noise rating in the tilde figure of merit shown earlier. The overall performance margin is therefore much lower due to the fact that the validated error floor consumes less than one-third of the 1 dB allocation margin for a typical topology. Therefore, there will be excess capacity remaining from a lower-level uncertainty (the major uncertainties associated with traffic allocation, implementation losses, and geographic precipitation).

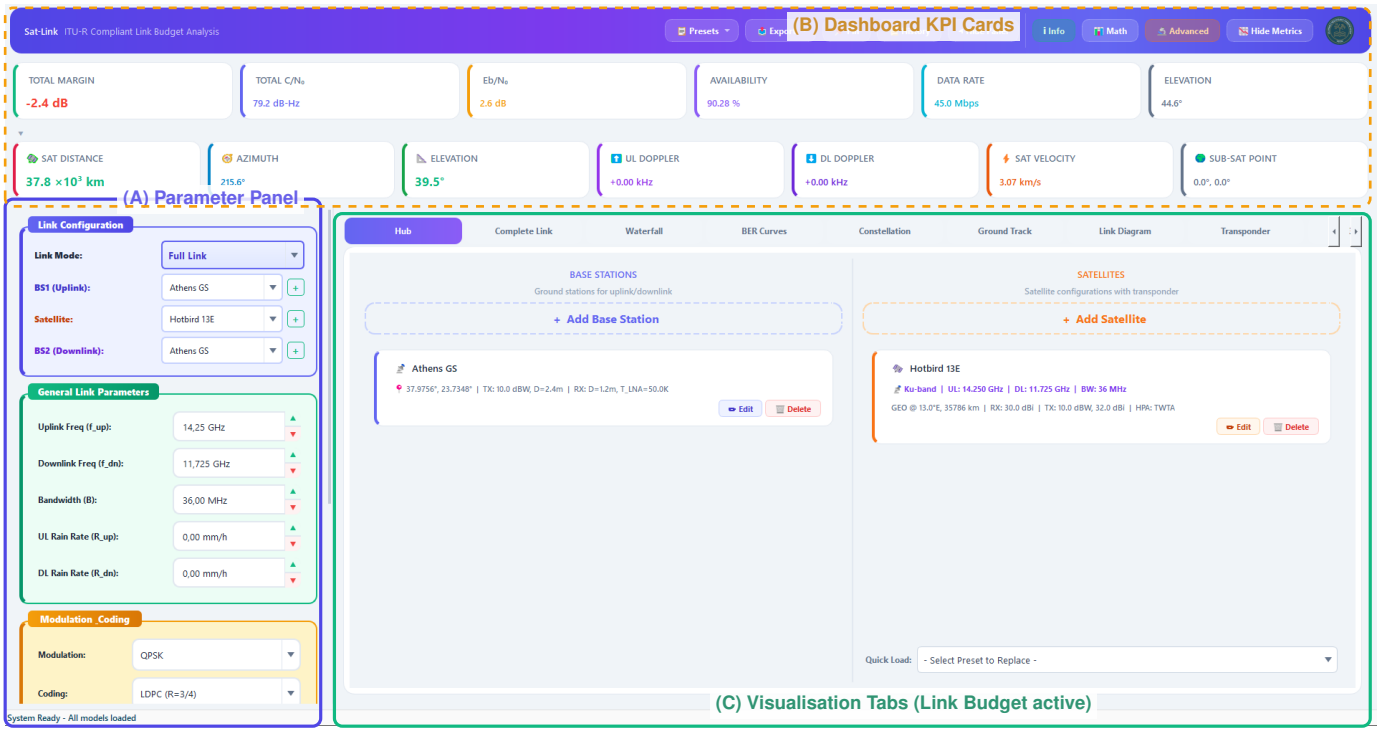

Fig. 4. Main application window with annotated interface regions. **(A)** Left parameter panel: all satellite, orbit, link, and ground-station configuration inputs. **(B)** Dashboard KPI cards: real-time link margin, C/N_0, and link-status indicators. **(C)** Seven-tab visualisation area with the Link Budget tab active, displaying the complete per-link budget breakdown.

TABLE I
VALIDATION AGAINST REFERENCE LINK BUDGETS

Parameter	GEO/Ku	LEO/S	GEO/Ka
Altitude (km)	35 786	550	35 786
Frequency (GHz)	12.0	2.2	20.0
EIRP (dBW)	52.0	25.0	58.0
G/T (dB/K)	14.2	10.5	18.7
Rain rate (mm/h)	0 (clear)	0	25
Results:			
C/N_0 — Ref. (dB-Hz)	87.3	73.1	78.5
C/N_0 — Sim. (dB-Hz)	87.1	73.3	78.2
Δ (dB)	-0.2	$+0.2$	-0.3

TABLE II
DERIVED ERROR METRICS PER VALIDATION SCENARIO

Metric	GEO/Ku	LEO/S	GEO/Ka
$\lvert\Delta C/N_0\rvert$ (dB)	0.2	0.2	0.3
Relative error (%)	0.23	0.27	0.38
Margin to ± 1 dB band (dB)	0.8	0.8	0.7

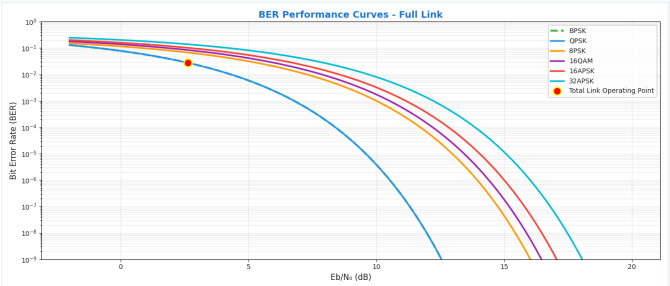

Fig. 5. BER performance curves for BPSK, QPSK, 8PSK, 16-QAM, 64-QAM, 16-APSK, and 32-APSK modulation schemes. The Shannon capacity limit is shown as a dashed line and the current operating point is marked.

C. Scope, Limitations, and Reproducibility

The validated core of Sat-Link is the deterministic link-budget chain (FSPL, atmospheric losses, transponder cascade, and C/N0 combination). Several advanced-analysis panels intentionally use reduced-order or heuristic relations (e.g., scenario deltas, synthetic time-series profiles, empirical IBO to C/IM mapping) to preserve interactivity. Therefore, the tool is positioned for education and early trade studies, not final regulatory filing or flight-qualification analysis.

Reproducibility is supported through the provided source code and pinned Python package list (`requirements.txt`); the paper figures were generated from the same codebase using Python 3.9 and PyQt5.

VIII. CONCLUSION

The study presented Sat-Link, a novel free open-source application for simulating satellite link budgets that adheres to the current ITU-R guidelines for educational purposes and preliminary trade studies. The functionality of this application includes performing validations against independent reference scenarios for demonstrating reliable technical consistency

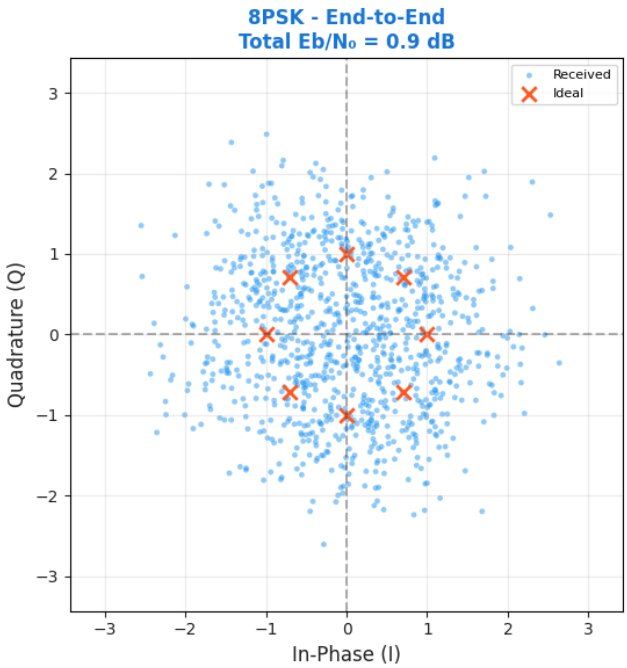

Fig. 6. IQ constellation diagram with Gaussian noise overlay at the current E_b/N_0 operating point. Gray-coded symbol mapping and decision boundaries are visible.

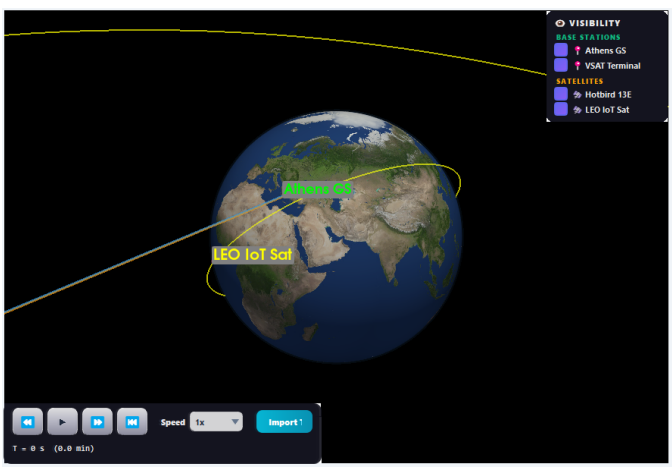

Fig. 7. Interactive 3D Earth view rendered with PyVista/VTK. The orbit path, satellite marker, and coverage footprint are displayed with full zoom and rotate controls.

within its intended use domain (i.e., <0.3 dB difference in required C/No - mean absolute error (MAE) = 0.23 dB). Thus, Sat-Link will facilitate the ability to conduct repeatable exercises in an academic environment while enabling rapid exploration of design spaces and avoiding reliance on opaque commercial processes. Future work includes calibration with actual measured data; development of larger-scale workflows for conducting interference coordination; conducting broader stress tests across various climates and bands; and furthering automated regression benchmarking techniques.

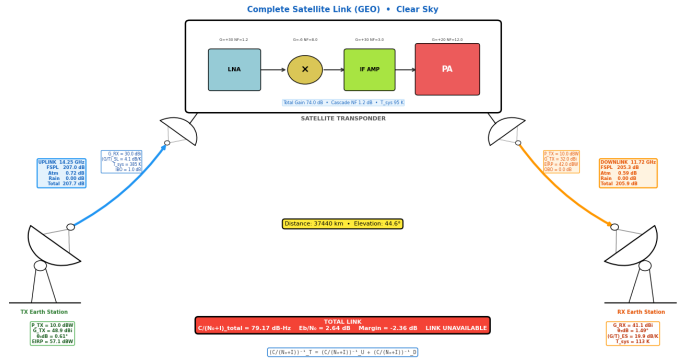

Fig. 8. Animated link diagram illustrating the satellite-to-ground communication path. Dynamic weather effects (clouds, rain drops) are toggled via a clear-sky/rain switch, simultaneously updating the budget calculations to reflect atmospheric impairments.

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References

[1] G. Maral, M. Bousquet, and Z. Sun, *Satellite Communications Systems: Systems, Techniques and Technology*, 6th ed. Wiley, 2020.
[2] ITU-R, "Satellite system design methodology," Rec. ITU-R S.1328-4, 2019.
[3] ETSI, "Digital Video Broadcasting (DVB); Second generation framing structure, channel coding and modulation systems for broadcasting, interactive services, news gathering and other broadband satellite applications," EN 302 307-1, V1.4.1, 2014.
[4] B. Sklar, *Digital Communications: Fundamentals and Applications*, 2nd ed. Prentice Hall, 2001.
[5] A. D. Usman, T. Afullo, A. J. Alonge, J. S. Mandeep, M. Ismail, and N. Roslee, "A comprehensive review of channel prediction models for satellite communications through rain in tropical regions: A case study of North Central Nigeria," *Heliyon*, vol. 5, no. 11, p. e03073, Nov. 2019, doi: 10.1016/j.heliyon.2019.e03073.
[6] ITU-R, "Propagation data and prediction methods required for the design of earth-space telecommunication systems," Rec. ITU-R P.618-14, 2023.
[7] ITU-R, "Specific attenuation model for rain for use in prediction methods," Rec. ITU-R P.838-3, 2005.
[8] ITU-R, "Attenuation by atmospheric gases and related effects," Rec. ITU-R P.676-12, 2019.
[9] ITU-R, "Attenuation due to clouds and fog," Rec. ITU-R P.840-8, 2019.
[10] ITU-R, "Reference radiation pattern for earth station antennas in the fixed-satellite service for use in coordination and interference assessment in the frequency range from 2 to 31 GHz," Rec. ITU-R S.465-6, 2010.
[11] ITU-R, "Reference FSS satellite antenna radiation pattern for use in interference assessment involving non-GSO satellites in the frequency range from 2 to 31 GHz," Rec. ITU-R S.1528, 2001.
[12] A. A. M. Saleh, "Frequency-independent and frequency-dependent nonlinear models of TWT amplifiers," *IEEE Trans. Commun.*, vol. 29, no. 11, pp. 1715–1720, 1981.
[13] C. Rapp, "Effects of HPA-nonlinearity on a 4-DPSK/OFDM-signal for a digital sound broadcasting system," in *Proc. ESA Workshop ESTEC*, Noordwijk, 1991, pp. 179–184.
[14] D. A. Vallado, *Fundamentals of Astrodynamics and Applications*, 4th ed. Microcosm Press, 2013.
[15] ITU-R, "Prediction method of fade dynamics on Earth-space paths," Rec. ITU-R P.1623-1, 2005.
[16] Ansys, "Systems Tool Kit (STK)." [Online]. Available: https://www.ansys.com/products/missions/ansys-stk